

5 CASE STUDY

In this section, we provide case studies of the Looped Neural Networks with different activation functions. We show that the Looped Neural Networks may have 2^d number of different fixed points.

First, we consider the polynomial activation function. We have the following robust fixed points statement.

Theorem 5.1 (A specific polynomial activation function with small perturbation). *Consider the L -layer looped neural networks $\text{NN}(x^{(0)}; W, b, L)$ defined in Definition 3.5. If the following conditions hold:*

- Let $C := \sqrt{\frac{15+\sqrt{65}}{8}}$ be a constant.
- Let $g(x) := -\frac{2}{5}x^4 + \frac{8}{5}Cx^3 + (\frac{3}{2} - \frac{12}{5}C^2)x^2 + (\frac{8}{5}C^3 - 3C)x + 1$ denote the polynomial activation function used in this looped neural networks.

Then there exists a set of noisy parameters W and b such that $x^{(t)} = \text{NN}(x^{(t-1)}; W, b, L) + h(x^{(t-1)}) \in \mathbb{R}^d$, for any $t \in [L]$, where the function h satisfies $|h(x)| \leq 1/m$ for every $x \in \mathbb{R}^d$ for some sufficiently large $m > d$. For this looped neural network, the following statements hold:

- The single layer of the looped neural network, $f(x; W, b)$, has at least 2^d robust fixed-points $p_1, \dots, p_{2^d}$ satisfying: For every $i \in [2^d]$, there exists a vector $\epsilon_i \in \mathbb{R}^d$, for any initial point $x^{(0)}$ with $\|x^{(0)} - p_i\|_\infty \leq \|\epsilon_i\|_\infty$, we have

$$\lim_{L \rightarrow \infty} \text{NN}(x^{(0)}; W, b, L) = p_i.$$

- For every $i \in [2^d]$, there exists a constant $c_i > 0$ and a constant $K_i \in [0, 0.9)$ such that for any $L \geq 2$, we have

$$\|\text{NN}(x^{(0)}; W, b, L) - p_i\|_\infty \leq K_i^L \cdot \|\epsilon_i\|_\infty + \frac{20}{m}.$$

Proof. Let $f(x; W, b)$ be a single layer version of $\text{NN}(x; W, b, 1)$. Let $m > d$ be a sufficiently large constant. Let $W = \frac{1}{m^2} \mathbf{1}_d \cdot \mathbf{1}_d^\top + (-\frac{1}{m^2} + 1) \text{diag}(\mathbf{1}_d) \in \mathbb{R}^{d \times d}$ denote the weight matrix. Let $b = [C, \dots, C]^\top \in \mathbb{R}^d$ denote the bias vector. Then it's clear that for each $j \in [d]$, we have

$$\begin{aligned} g(\langle w_j, x \rangle + b_j) &= -\frac{2}{5}x_j^4 + \frac{3}{2}x_j^2 + \frac{1}{m^2}h_j(x) \\ &= f_j(x; W, b) + \frac{1}{m^2}h_j(x) \end{aligned}$$

where $h_j(x)$ is a polynomial in $x_1, \dots, x_d$. For every $j \in [d]$, there exists a $D_1 = (1.302, 1.502)$ and $D_2 = (-0.3, 0.3)$, it satisfies that $f_j(x; W, b) \in D_1$ when $x_j \in D_1$ and $f_j(x; W, b) \in D_2$ when $x_j \in D_2$. By the proof in Lemma B.1, there exists a $K_j = 0.92$ such that for any $x_j \in D_1 \cup D_2$ we have

$$|f'_j(x; W, b)| \leq K_j.$$

It's obvious that when m is sufficiently large and $\|\epsilon_i\|_\infty$ is small enough, we will have $|\frac{1}{m^2}h_j(x)| \leq \frac{1}{m}$, where we can see $\frac{1}{m^2}h_j(x)$ as $h(x)$ in statement of Theorem 4.2. Then by combining the result of

Theorem 4.2, we have that for each dimension $j \in [d]$, we have 2 robust fixed points. So we can get $f(x; W, b)$ has 2^d Robust Fixed Points trivially. And for any $i \in [2^d], j \in [d]$, we have

$$\begin{aligned} & |\text{NN}_j(x^{(t)}; W, b, L) - p_{i,j}| \\ & \leq K_{i,j}^t |\text{NN}_j(x^{(0)}; W, b, L) - p_{i,j}| + \frac{20}{m} \\ & \leq K_{i,j}^t \epsilon_{i,j} + \frac{20}{m} \end{aligned}$$

where the first step comes from the result of Theorem 4.2, the second step comes from the range of values for the initial point. Then use the definition of $\|\cdot\|_\infty$, we have

$$\|\text{NN}(x^{(t)}; W, b, L) - p_i\|_\infty \leq K_i^t \cdot \|\epsilon_i\|_\infty + \frac{20}{m}.$$

Then we complete the proof. $\square$

In Theorem 5.1, we can see that our Looped Neural Network has 2^d different robust fixed points solutions. Recall that the previous analysis tools can only handle single fixed point analysis.

We can show the existence of 2^d different robust fixed points solutions when the Looped Neural Network uses exponential activation as well.

Theorem 5.2 (A specific exponential activation function with small perturbation). *Consider the L -layer looped neural networks $\text{NN}(x^{(0)}; W, b, L)$ defined in Definition 3.5. If the following conditions hold:*

- Let $C := -2.15$ be a constant.
- Let $g(x) := \exp(x^3 + (-2 - 3C)x^2 + (3C^2 + 4C)x + \ln 2) - 1$ denote the exponential activation function used in this looped neural networks.

Then there exists a set of noisy parameters W and b such that $x^{(t)} = \text{NN}(x^{(t-1)}; W, b, L) + h(x^{(t-1)}) \in \mathbb{R}^d$, for any $t \in [L]$, where the function h satisfies $|h(x)| \leq 1/m$ for every $x \in \mathbb{R}^d$ for some sufficiently large $m > d$. For this looped neural network, the following statements hold:

- The single layer of the looped neural network, $f(x; W, b)$, has at least 2^d robust fixed-points $p_1, \dots, p_{2^d}$ satisfying: For every $i \in [2^d]$, there exists a vector $\epsilon_i \in \mathbb{R}^d$, for any initial point $x^{(0)}$ with $\|x^{(0)} - p_i\|_\infty \leq \|\epsilon_i\|_\infty$, we have

$$\lim_{L \rightarrow \infty} \text{NN}(x^{(0)}; W, b, L) = p_i.$$

- For every $i \in [2^d]$, there exists a constant $c_i > 0$ and a constant $K_i \in [0, 0.9)$ such that for any $L \geq 2$, we have

$$\|\text{NN}(x^{(0)}; W, b, L) - p_i\|_\infty \leq K_i^L \cdot \|\epsilon_i\|_\infty + \frac{20}{m}.$$

Proof. Let $f(x; W, b)$ be a single layer version of $\text{NN}(x; W, b, 1)$. Let $m > d$ be a sufficiently large constant. Let $W = \frac{1}{m^2} \mathbf{1}_d \cdot \mathbf{1}_d^\top + (-\frac{1}{m^2} + 1) \text{diag}(\mathbf{1}_d) \in \mathbb{R}^{d \times d}$ denote the weight matrix. Let $b = [C, \dots, C]^\top \in \mathbb{R}^d$ denote the bias vector. Then it's clear that for each $j \in [d]$, we have

$$g(\langle w_j, x \rangle + b_j) = \exp(x_j^3 - 2x_j) - 1 + \frac{1}{m^2} h_j(x)$$

where the first step follows from $\exp(z) = 1 + O(z)$ when $|z| \leq 1$ and $|x_j| \leq 1$. $h_j(x)$ is an exponential function in $x_1, \dots, x_d$. For every $j \in [d]$, there exists a $D_1 = (-0.1, 0.1)$ and $D_2 = (-1.01, -0.81)$, it satisfies that $f_j(x; W, b) \in D_1$ when $x_j \in D_1$ and $f_j(x; W, b) \in D_2$ when $x_j \in D_2$. By the proof in Lemma C.1, there exists a $K_j = 0.85$ such that for any $x_j \in D_1 \cup D_2$ we have

$$|f'_j(x; W, b)| \leq K_j.$$

Then we finish the proof by following the same proof statement in Theorem 5.1. □

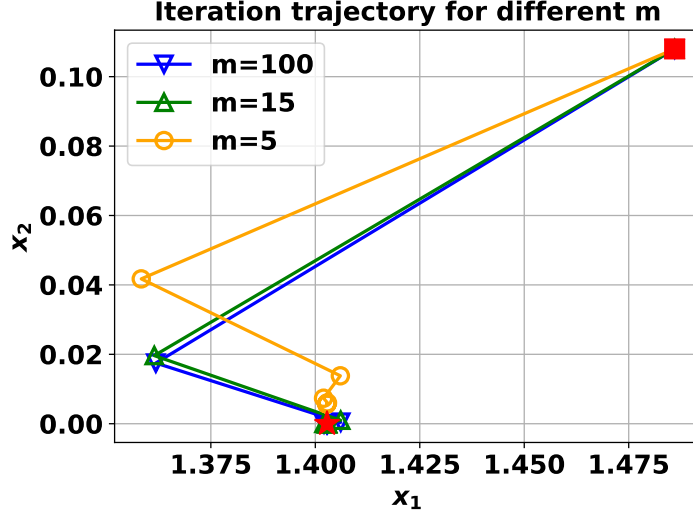


Figure 3: Different iteration trajectory when taking different m in Theorem 4.2. We consider the 2- d case, where the two axes are x_1 and x_2 . The red square (■) in this figure is the initial point $x^{(0)} \in \mathbb{R}^2$. The red star (★) in this figure is the robust fixed point without noise. The m represents the inverse of the noise level, i.e., noise = $1/m$. From the figure, we can see that even though the noise changes the fixed point iteration process, different trajectories will eventually converge to the robust fixed point.

6 EXPERIMENTS

6.1 Setup

In our experiments, we used Python 3.8 for simulation and employed Matplotlib 3.4.3 library to visualize the experimental results. We conduct three simulations in our work:

- We are going to verify the polynomial function in Lemma B.1 and the exponential function in Lemma C.1 actually have 2 fixed point.
- We are going to verify that the neural network in Theorem 5.1 has 2 fixed point in each dimension.
- Recall that m represents the inverse of the noise level in Theorem 5.1. We are going to verify that even the noise changes in the neural network of Theorem 5.1, different fixed point iteration trajectories will eventually converge to the Robust Fixed Point.

6.2 Results

In Figure 1, we simulate the Banach fixed point iteration on a polynomial function $f(x) = -\frac{2}{5}x^4 + \frac{3}{2}x^2$ and an exponential function $f(x) = \exp(x^3 - 3x^2) - 1$. It shows that there are two fixed points ($x_1 = 0$ and $x_2 \approx 1.403$) of the polynomial function and the points in the neighborhood of a fixed point will iterate to the corresponding fixed point through fixed-point iteration. It also shows that there are two fixed points ($x_1 = 0$ and $x_2 \approx -0.910$) of the exponential function, and the points in the neighborhood of a fixed point will iterate to the corresponding fixed point through fixed-point iteration.

In Figure 2, we simulate a 2- d neural network in Theorem 5.1. It shows that in the first dimension of the neural network, we have two robust fixed points ($x_1 = 0$ and $x_1 \approx 1.403$), and in the second dimension of the neural network, we have two robust fixed points ($x_2 = 0$ and $x_2 \approx 1.403$).

In Figure 3, we simulate the effect of different values of m on the fixed-point iteration. It shows that even though the noise changes the fixed point iteration process, different trajectories, $m = 5$, $m = 15$, and $m = 100$, when taking the same initial point (■), will eventually converge to the robust fixed point (★).

7 CONCLUSION

This study provides an analytical framework for understanding fixed point iterations in deep neural networks. We established theorems for multiple fixed points and their robustness under noise. Case studies with polynomial and exponential activation functions show the looped neural networks may have an exponential number of robust fixed points, demonstrating our approach’s effectiveness. Our findings offer new insights into deep neural networks, contributing to the analysis toolkit and potentially enhancing deep neural network understanding and efficient algorithm design.

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